



Transparency between Research Methods and Results

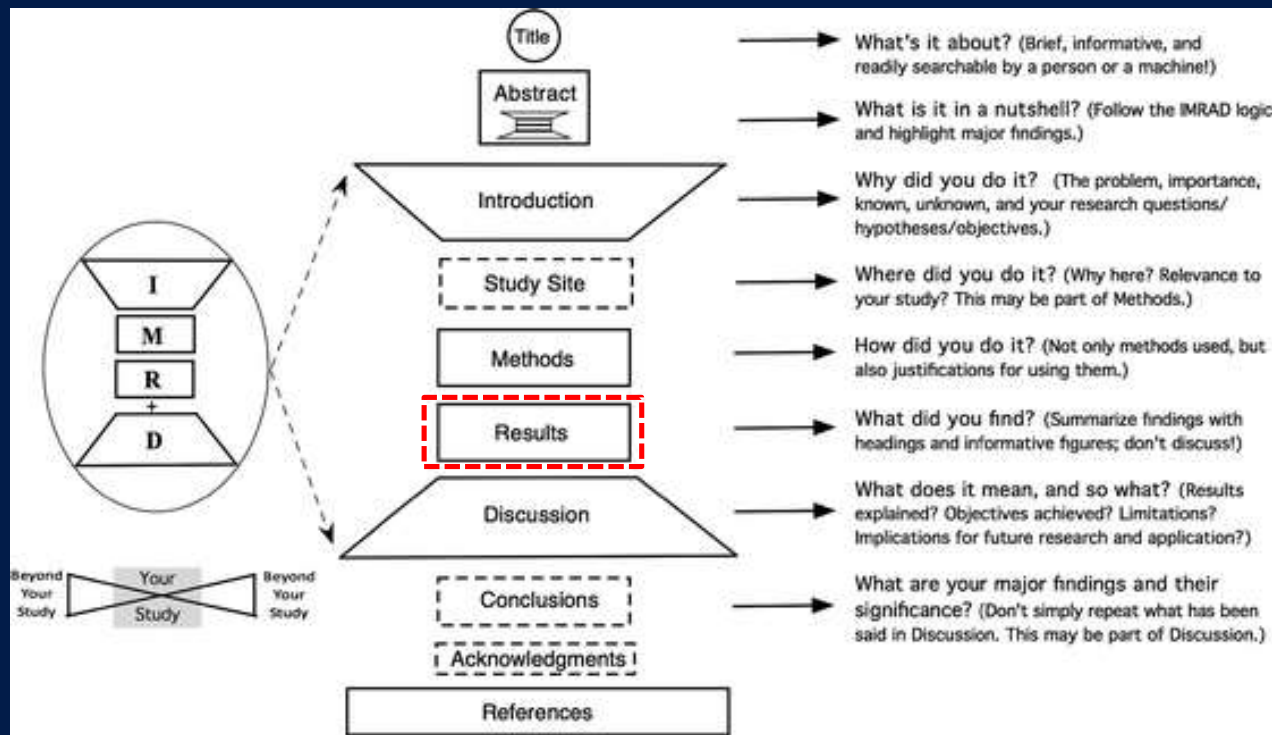
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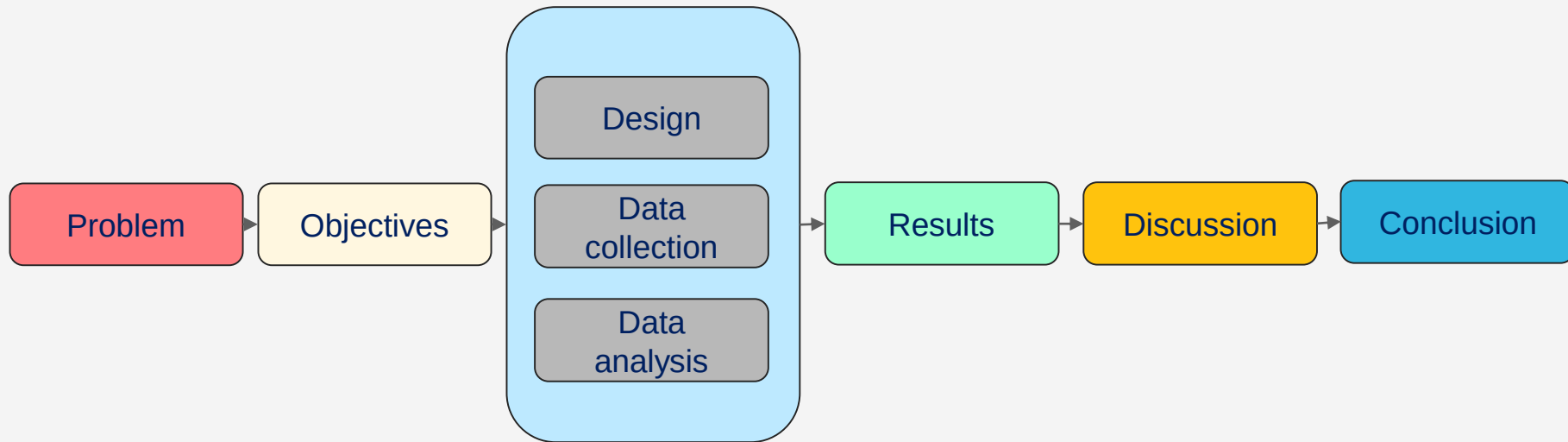
Outline

- Defining transparency in research
- Linking methods to results
- Reporting quantitative and qualitative results transparently

Recap on IMRAD method



Research Workflow



Transparency

methodological transparency refers to “the degree of detail and disclosure about the specific steps, decisions, and judgment calls made during a scientific study”

Transparency

Entails clear reporting of design,
sampling, analysis, uncertainty,
limitation and ethics

Benefits of research transparency

- Credibility
- Reproducibility
- Scientific Integrity
- Trustworthiness

Objectives → Methods → Results

Objective	Data Collection	Analysis	Results
To determine flood exposure	Household survey	Descriptive statistics	Exposure percentages
To understand community perceptions	Interviews	Thematic analysis	Themes
To predict flood risk	GIS & Remote Sensing	Random Forest	Flood risk maps

Discuss more examples from participants

Example (identify gap, objectives and methods)

To address food security needs of the ASAL areas in Kenya, the government introduced a new Water Management Masterplan that gives more freedom to large scale agricultural users on the use of groundwater. In particular, the master plan promotes extraction of underground water for use in large scale agricultural irrigation at a marginal fee (0.5 USD per cubic meter). While this has been appreciated by the farming communities, there is observable change in the general vegetation health with persistent drying of indigenous trees and herbs, affecting the livelihoods of pastoralist communities. **Information on the link between underground water extraction and vegetation degradation is scarce.** The overall aim of the current study is to assess the link between ground water extraction and changes in vegetation condition. In particular, satellite-derived data from Google Earth Engine (GEE) was used to quantify the correlation between monthly groundwater volume and enhanced vegetation index (EVI) at 117 randomly sampled locations across the country. A linear regression model of EVI from groundwater showed a moderate positive link between changes in vegetation condition and groundwater (R-square value of 0.58, p-value < 0.001). These results show that extraction groundwater could be one of the drivers of vegetation change. Further assessment with additional variables including land use change practices and the influence of long dry seasons should be considered.

Identifying gap, objectives, methods and potential results

IMRAD ELEMENT	Description
GAP	Lack of empirical evidence on the link between underground water dynamics and changes in vegetation Condition
OBJECTIVE	Assess the link between vegetation condition and groundwater extraction
METHOD	Data source: Google Earth Engine (groundwater and EVI) Sampling: Random Sampling at 117 Analysis: Linear regression – Correlation, R-square, equation Tools: Excel
RESULT	R-square value Plot a chart of visualization

Documenting Results in Detail, and linking to methods

To assess the link between vegetation condition and groundwater, a linear regression model between EVI and groundwater was used. The results showed a moderate positive correlation between EVI and groundwater with an R-square value of 0.57. The model also showed that changes in EVI can be predicted from groundwater using the Equation 1

$$\text{EVI} = 0.0003\text{Groundwater} + 0.1223$$

Equation 1.

Figure 1 represents the linear relationship between EVI and groundwater, showing a positive link (dotted line) between the two variables.

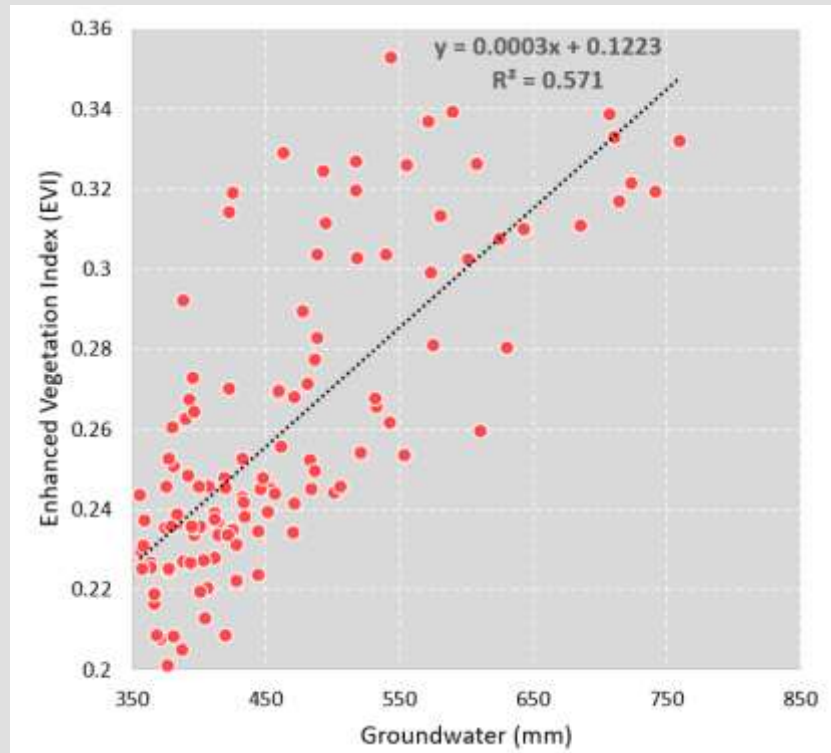


Figure 1: Linear relationship between enhanced vegetation index (EVI) and groundwater

**No results should be
reported without
corresponding method**

Reporting results from quantitative methods

Present findings in the same order as described in Methods.

Descriptive Statistics

- Means
- Standard deviations
- Frequencies
- Percentages

Inferential Statistics

Include

- Test statistic
- Degrees of freedom
- p-values
- Confidence intervals
- Effect size

Tables, Figures and Maps

Good visualizations should include:

- Clear titles
- Units
- Legends
- Sample size
- Source
- Numbering

Using Charts to report results

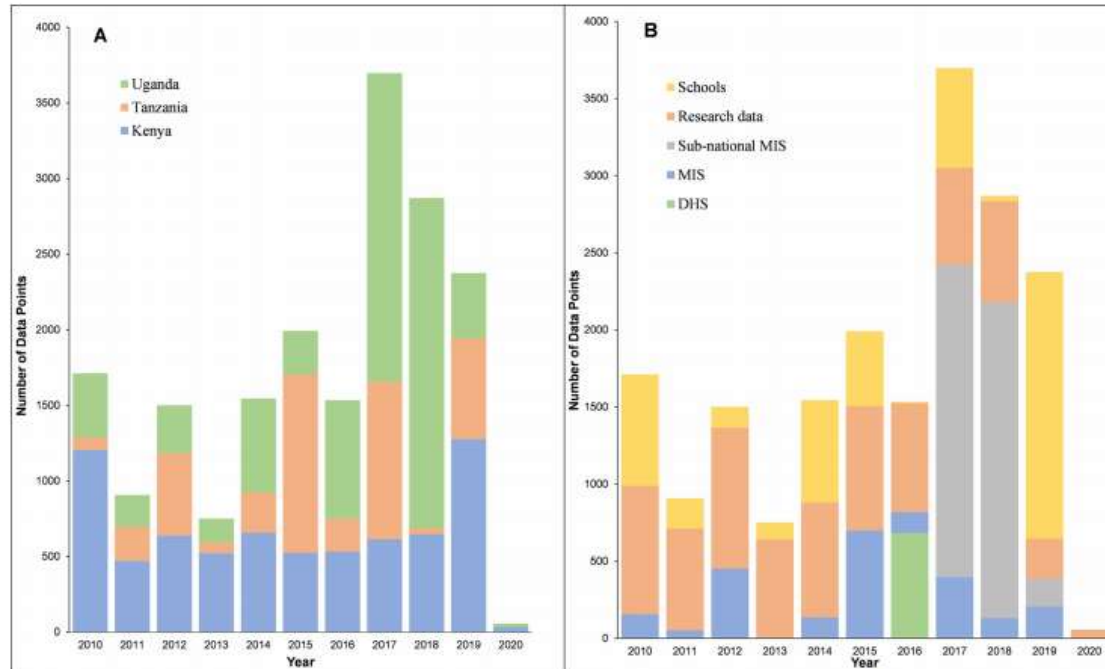


Fig 2. Assembled surveys by year and data source. (A) Temporal distribution of surveys 2010–2020 by country (B) Temporal distribution of surveys 2010–2020 according to the data source

Reporting results from qualitative methods

Present findings by theme

Theme 1: Limited Access to Climate Information

Quotation

"Flood warnings usually arrive after the flooding has already started."

Interpretation

Delayed communication increases household vulnerability.

Reporting results from mixed methods

Integrate quantitative and qualitative findings

Quantitative	Qualitative	Interpretation
72% flooded	Residents cited blocked drainage	Infrastructure contributes to flood risk
63% lacked insurance	Participants reported poverty	Financial vulnerability limits resilience

Reporting results from AI and GIS

For GIS studies report

- Spatial resolution
- Coordinate system
- Classification algorithm
- Accuracy assessment
- Validation

For AI studies report

- Model used
- Hyperparameters
- Feature selection
- Accuracy
- Precision
- Recall
- F1-score
- Explainability

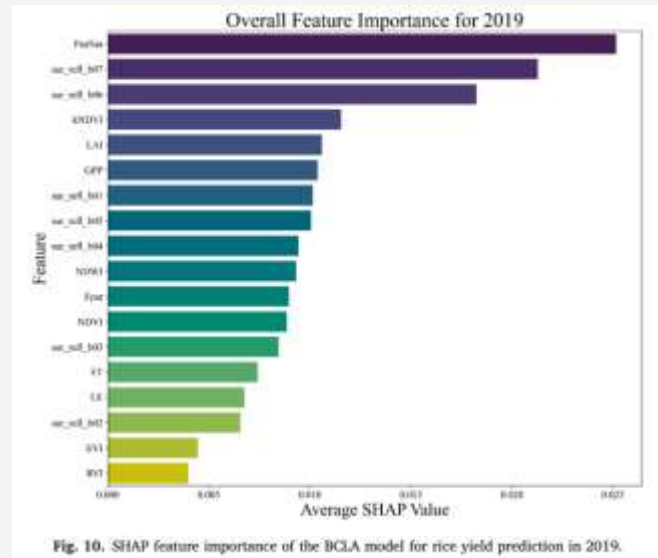


Fig. 10. SHAP feature importance of the BCLA model for rice yield prediction in 2019.

Mistakes

- Reporting results not described in Methods
- Introducing new variables
- Mixing Results with Discussion
- Too many quotations without analysis
- Poor figure captions
- Lack of uncertainty reporting

Best Practices

- Every objective should have a corresponding method and results
- Results should be presented in the same order of the methods
- Use tables and figures effectively
- Report uncertainty and limitations of the study

Questions and discussions